

S&S QUIZ-1 SOLUTION

Sol 1(a):- For Periodic signal,

$$x(t+T) = x(t) \quad \forall t$$

where, $T = \text{FTP (smallest positive)}$

Given, $x(t) = e^{j(\pi t - 1)}$ — (1)

$$\begin{aligned} x(t+T) &= e^{j[\pi(t+T) - 1]} \\ &= e^{j(\pi t - 1)} \cdot e^{j\pi T} \end{aligned} \quad \text{--- (2)}$$

In eqⁿ(2), if $T=2$ then — $e^{j\pi T} = e^{j2\pi} = 1$

Hence, $x(t+T) = x(t)$

$\therefore x(t)$ is periodic signal with FTP, $T=2 \rightarrow (2 \text{ MARK})$

Sol 1(b):- For Periodic signal,

$$x[n+N] = x[n] \quad \forall n$$

where, $N = \text{FTP} \in \mathbb{I}$ & smallest value

Given, $x[n] = \cos\left(\frac{\pi}{8}n^2\right)$ — (1)

$$\begin{aligned} x[n+N] &= \cos\left[\frac{\pi}{8}(n+N)^2\right] \\ &= \cos\left[\frac{\pi}{8}n^2 + \frac{\pi}{8}N^2 + \frac{2\pi}{8}nN\right] \end{aligned} \quad \text{--- (2)}$$

As we know that — $\cos(2\pi \pm \theta) = \cos \theta$

In eqⁿ(2), $\frac{\pi}{8}N^2 + \frac{2\pi}{8}nN = 2\pi k$

smallest integer value for which above condition satisfy is — $N=8$

Hence, $x[n+N] = x[n]$

$\therefore x[n]$ is periodic signal with FTP, $N=8 \rightarrow (2 \text{ MARK})$

Sol 2(a)÷ Given that-

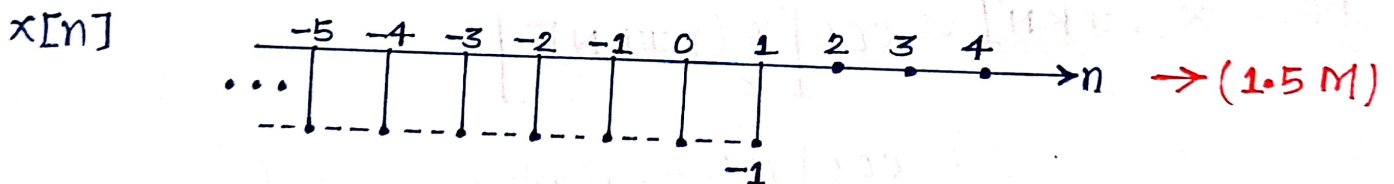
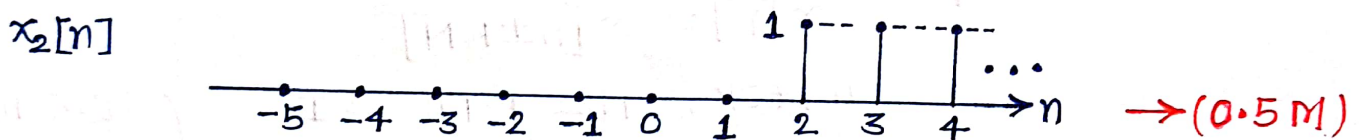
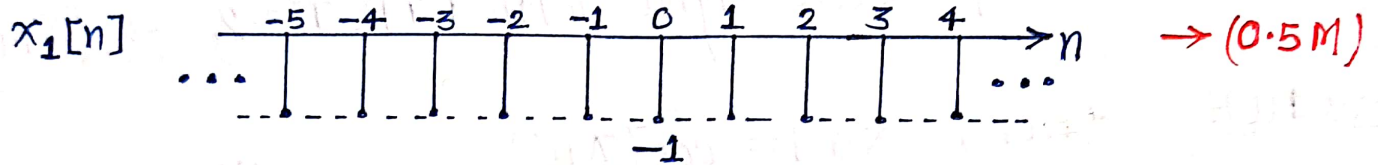
$$x[n] = -1 + \sum_{k=2}^{\infty} \delta[n-k]$$

$$= x_1[n] + x_2[n]$$

where,

$$x_1[n] = -1$$

$$x_2[n] = \sum_{k=2}^{\infty} \delta[n-k]$$



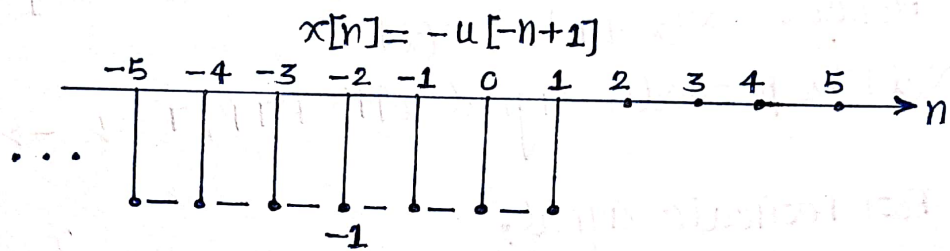
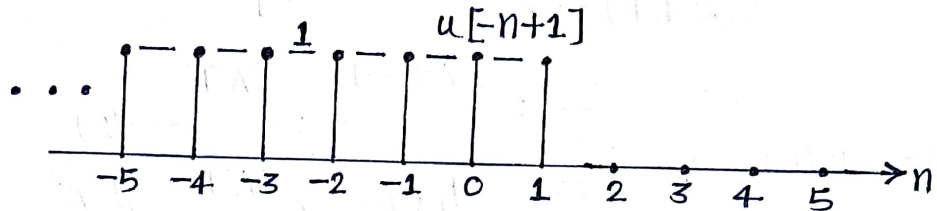
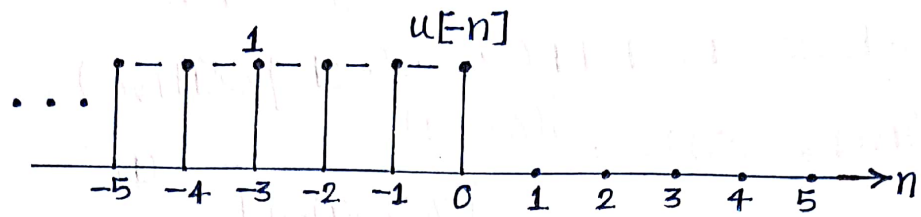
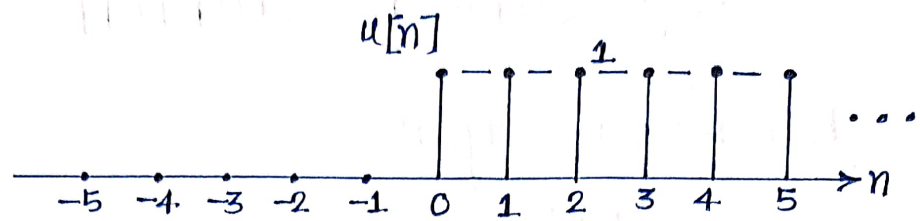
Sol 2(b)÷ By the above plot of $x[n]$, we can get -

$$x[n] = \begin{cases} -1, & n \leq 1 \\ 0, & n \geq 2 \end{cases}$$

$$= -u[-(n-1)]$$

$$= -u[-n+1] = Au[Mn+n_0]$$

Hence, $A = -1$, $M = -1$, $n_0 = 1$ → (3 X 0.5 Mark)



$\rightarrow (4 \times 0.25 \text{ Mook})$